

Kingdom of Science: engineering.pdf

Charcoal Kiln

Pyrolysis decomposes wood in low oxygen to yield carbon charcoal. Required: Hardwood logs, clay dome, turf sods. Procedure: Stack logs in cone, cover with 15cm clay mud, ignite top chimney, restrict base vents. Seal completely when smoke turns clear. Safety: Cool 48h before opening.

Lime Mortar & Cement

Slaked lime reacts with CO_2 to form calcium carbonate stone matrix. Required: Slaked lime, clean sand, crushed brick dust. Procedure: Mix 1 part slaked lime, 3 parts sand, 0.5 parts brick dust with water to stiff paste. Apply between stone blocks. Safety: Wear gloves.